

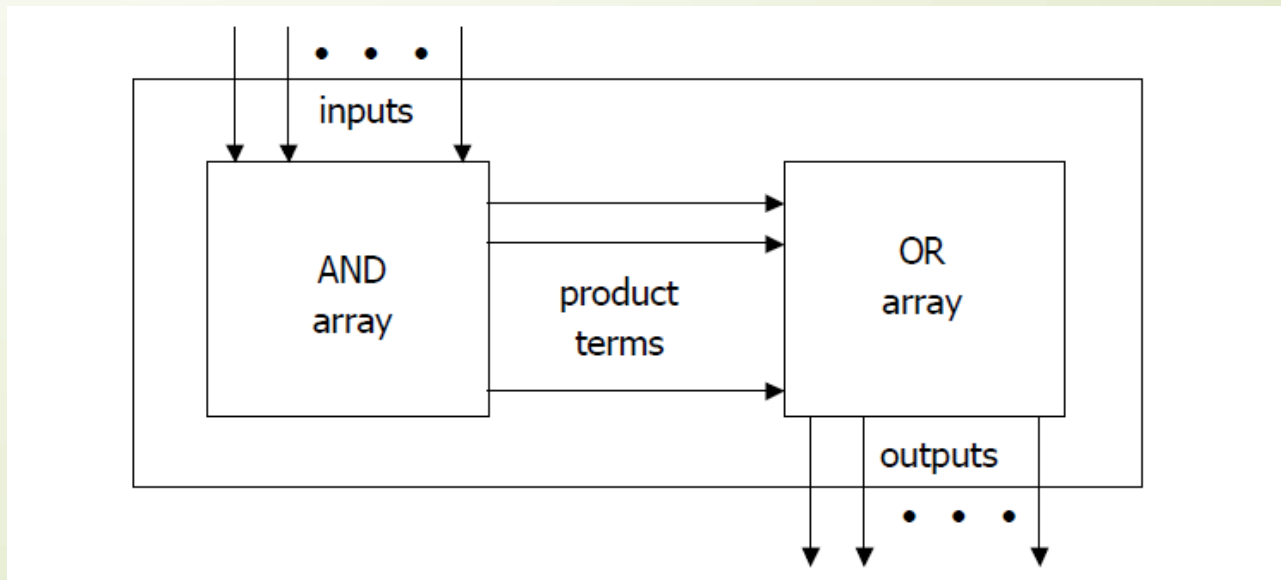


CDA 3103

Recitation 4 – Logic Design

Programmable Logic Array (PLA)

- An implementation of sum of products
- A structured-logic element composed of a set of inputs and their complements and also 2 stages of logic:
 - The first generates product terms of inputs and input complements
 - The second generates sum terms of the product terms (minterms)

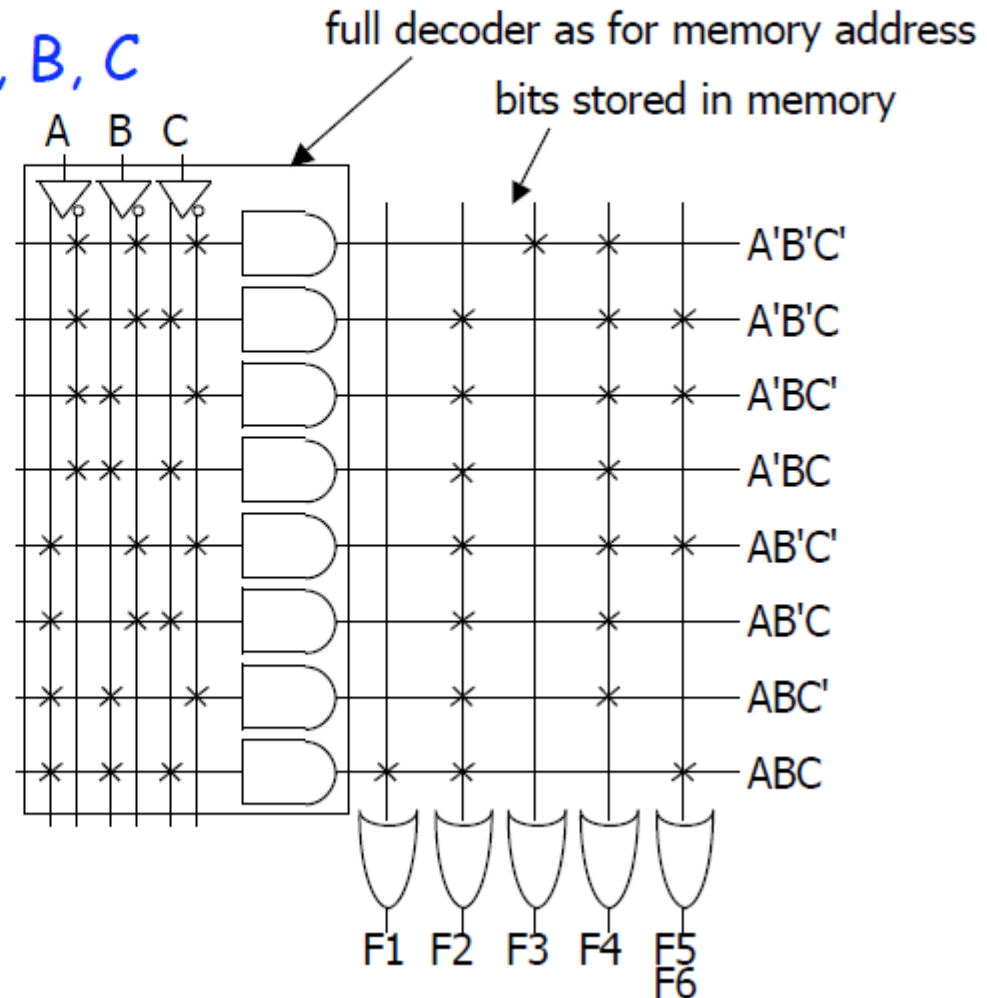


Programmable Logic Array Example

■ Multiple functions of A, B, C

- $F1 = A B C$
- $F2 = A + B + C$
- $F3 = A' B' C'$
- $F4 = A' + B' + C'$
- $F5 = A \text{ xor } B \text{ xor } C$
- $F6 = (A \text{ xnor } B \text{ xnor } C)'$

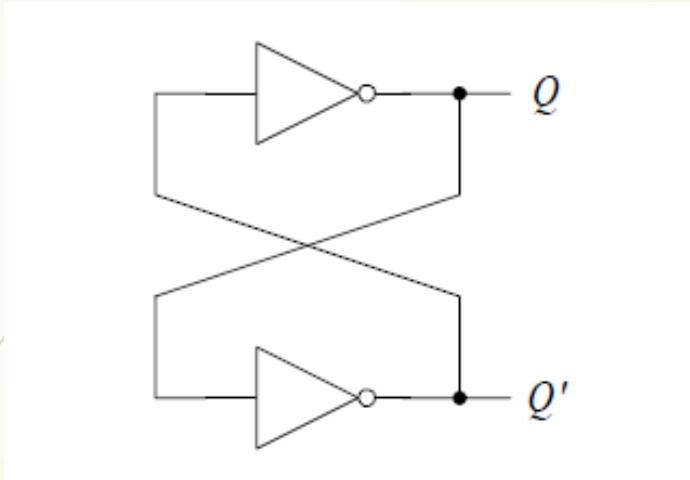
A	B	C	F1	F2	F3	F4	F5	F6
0	0	0	0	0	1	1	0	0
0	0	1	0	1	0	1	1	1
0	1	0	0	1	0	1	1	1
0	1	1	0	1	0	1	0	0
1	0	0	0	1	0	1	1	1
1	0	1	0	1	0	1	0	0
1	1	0	0	1	0	1	0	0
1	1	1	1	1	0	0	1	1



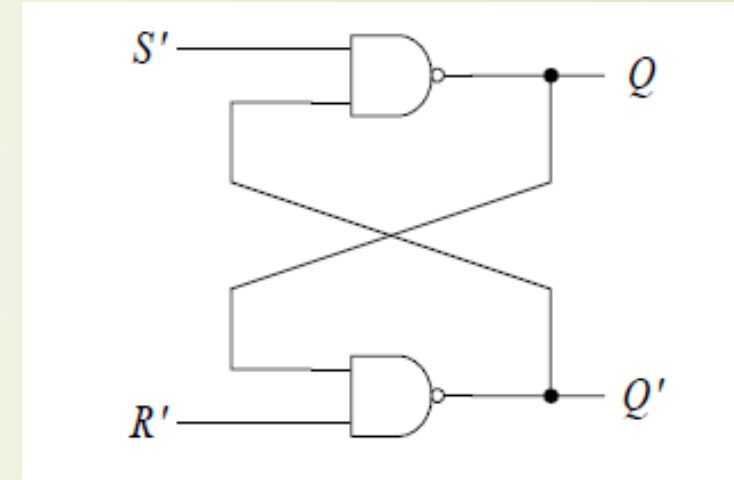


Combinational Logic Circuits Vs Sequential Logic Circuits

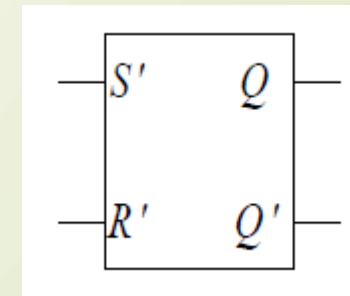
- ▶ **Combinational logic** refers to circuits whose output is a function of the *present value of the inputs* only.
 - ▶ As soon as inputs are changed, the information about the previous inputs is lost, that is, combinational logic circuits have no memory
- ▶ **Sequential logic** circuits are those whose outputs are also dependent upon past inputs, and hence outputs. In other words the output of a sequential circuit may depend upon its previous outputs and so in effect has some form of "memory", such as bistables or latches.
 - ▶ **Finite state machine (FSM)** is the mathematical model of a sequential circuit which includes states, inputs, and outputs.



Bistable element
 Simplest sequential circuit
 Two inverters
 No inputs
 Cannot change outputs Q and Q'
 If $Q=0$, $Q'=1$
 If $Q=1$, $Q'=0$
 Stable situation: 2 stable states

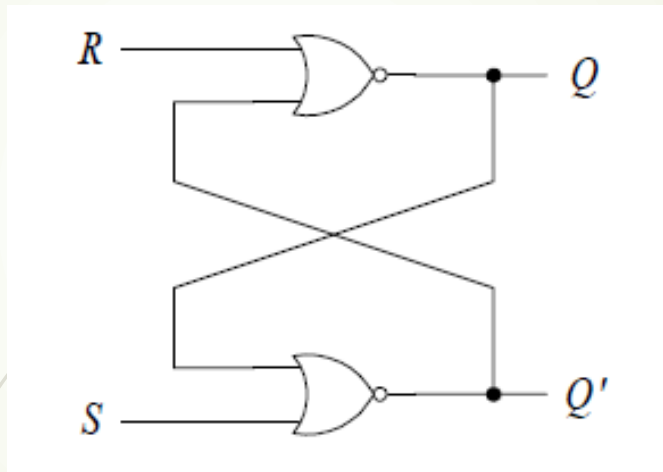


SR latch circuit using NAND gates

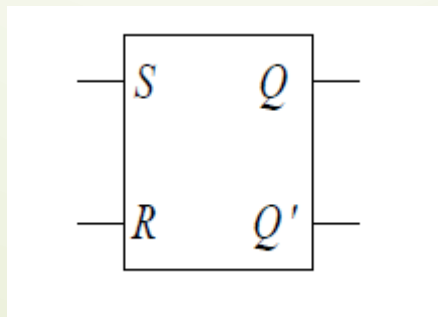


Logic symbol

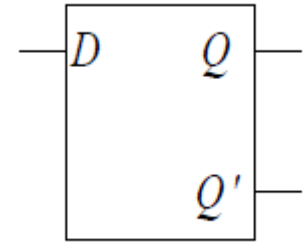
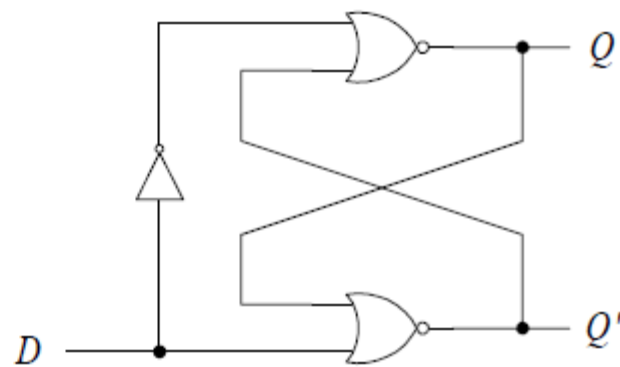
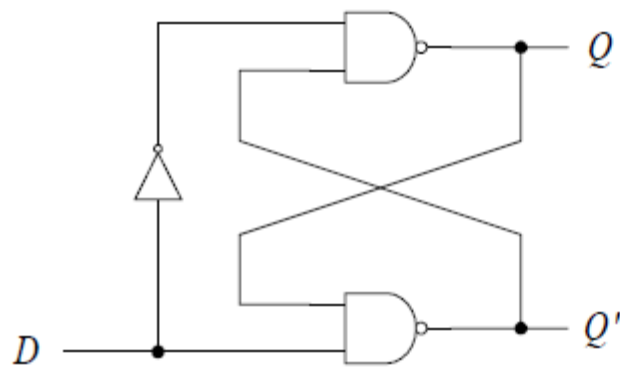
2 inputs S' and R'
 Primes denote inputs are active low
 If $S'=0$, $Q=1$ (Latch is Set)
 And then if $R'=1$, $Q'=0$



SR Latch using NOR gates



Logic symbol

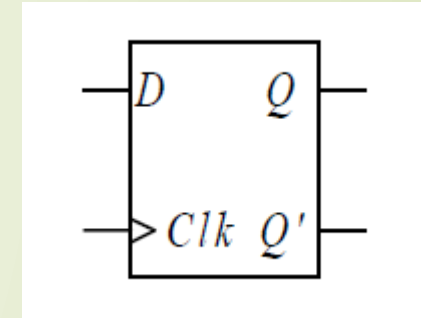
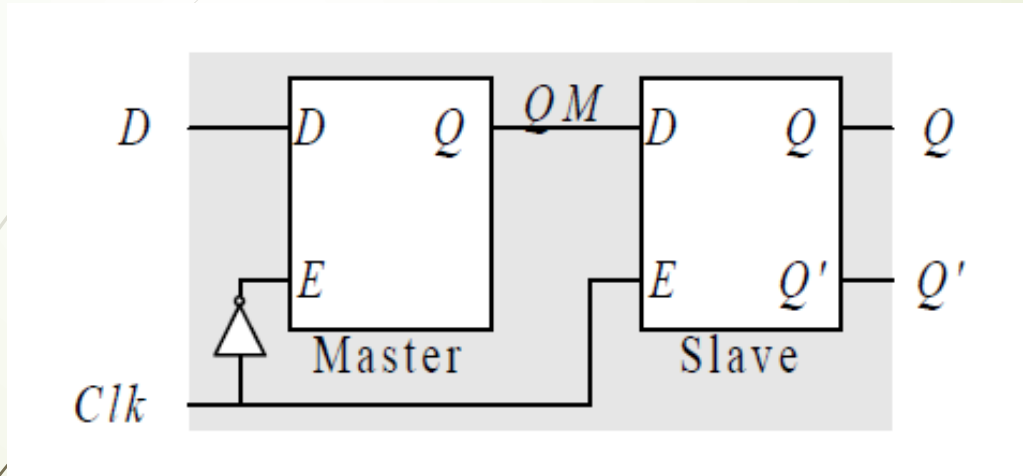


D Latch

An inverter, an input D ,
and 2 outputs Q , Q'

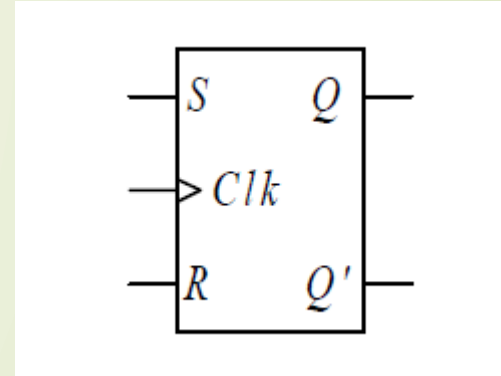
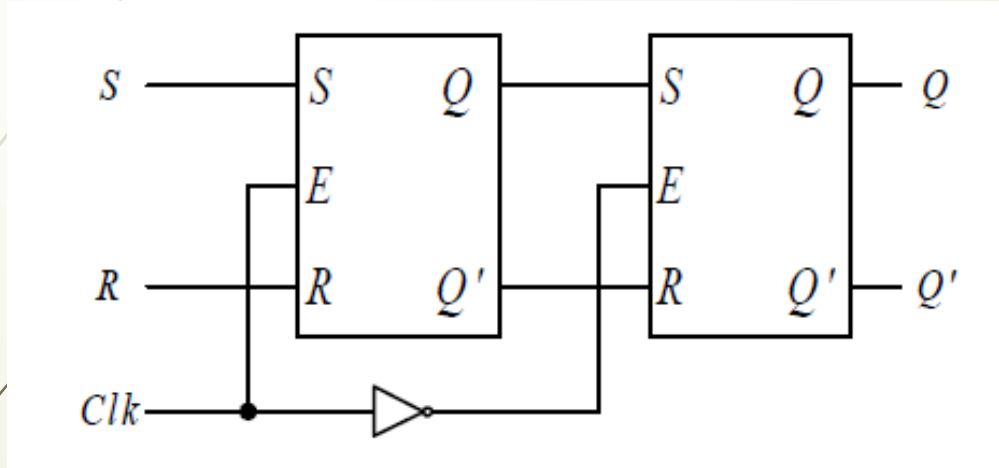
- P.S.: Latches are called level-sensitive because their output follows their inputs as long as they are enabled.
- Sometimes it is more useful to have the output change only at the rising or falling edge of the signal. This signal is clock signal → **D Flip-Flop**

D Flip-Flop



- An edge triggered flip-flop where 2 D-latches and 1 clock are connected
- First latch is called master latch, enabled when $Clk=0$
- When $Clk=1$, master is disabled and slave is enabled so that output from the master is transferred to the slave
- All state changes occur on a clock edge
- Either rising or falling edge is active and causes state changes to occur

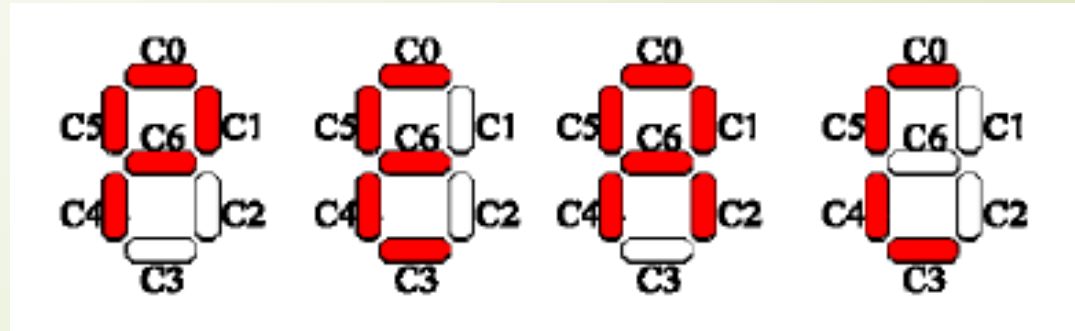
SR Flip-Flop



An SR flip-flop with 2 SR-latches and a clock connected
Useful in control applications where we want to be able to Set or Reset the data bit
Unlike SR-latches, SR flip-flops change their content only at the active edge of the clock signal.

Combinational Circuits Example

- Design a logic network that can display the characters in "PEACE" on a seven-segment display unit. Write the truth table for the output functions C0-C6, minimize them by using a Karnaugh Map, and implement using discrete logic gates.



- Character "A" will be displayed if segments (C0, C1, C2, C4, C5, C6) of a 7-segment display are turned on and segment (C3) is turned off. Since only 4 characters will be displayed, input of 2 bits to encode them will be needed for the given design. Call those inputs X0 and X1.

Assign the four different characters to a 2-bit code:

P 00

E 01

A 10

C 11

Construct a truth table:

X	Y	C0	C1	C2	C3	C4	C5	C6
0	0	1	1	0	0	1	1	1
0	1	1	0	0	1	1	1	1
1	0	1	1	1	0	1	1	1
1	1	1	0	0	1	1	1	0

Note which output functions are identical:

$$C0 = C4 = C5$$

Minimize output with karnaugh maps:

$$C0 = C4 = C5 = 1$$

1	1
1	1

$$\begin{aligned} X'Y' + XY' + X'Y + XY &= \\ Y'(X+X') + Y(X'+X) &= \\ Y' + Y &= 1 \end{aligned}$$

$$C1 = Y'$$

1	1
0	0

$$C2 = XY'$$

0	1
0	0

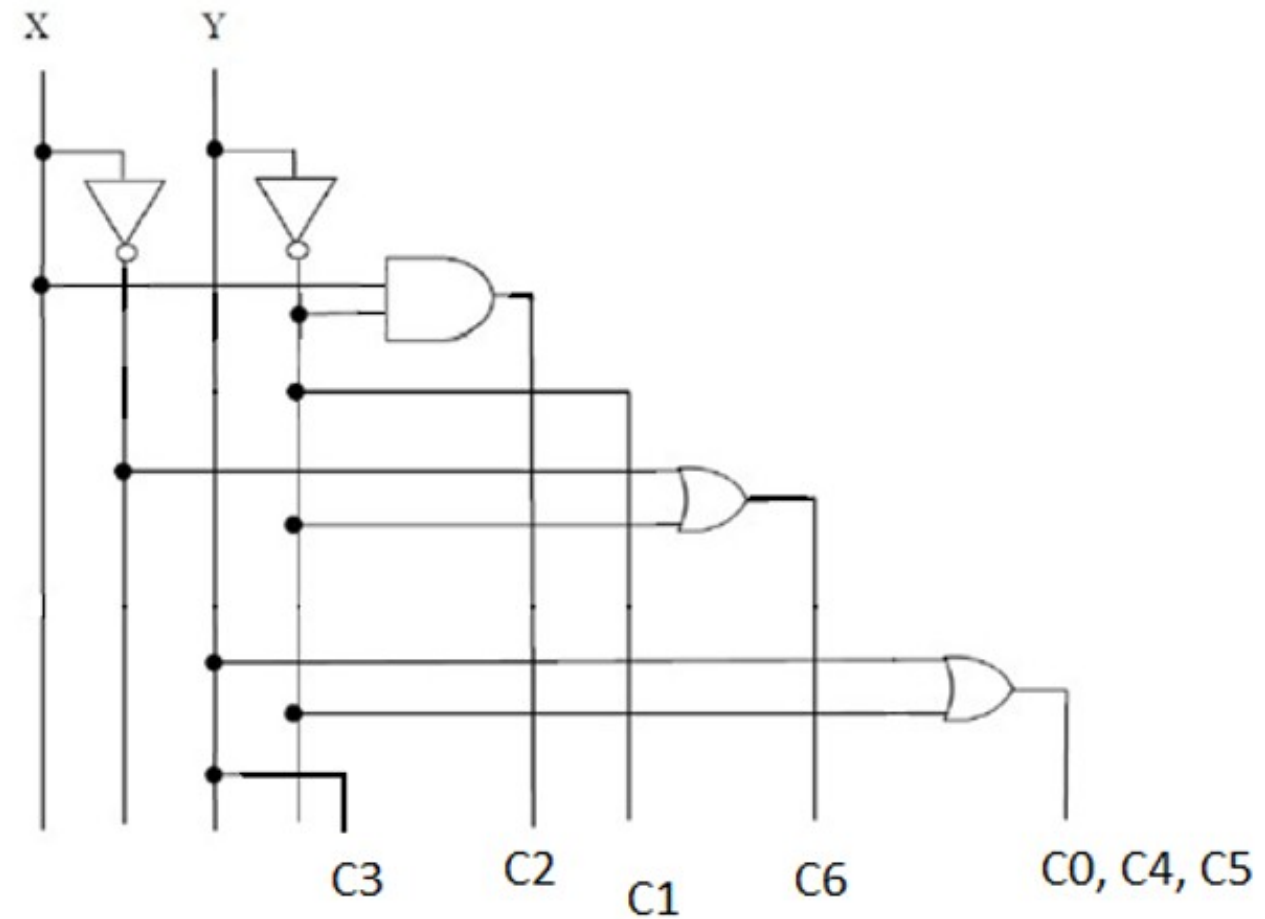
$$C3 = Y$$

0	0
1	1

$$C6 = X' + Y'$$

1	1
1	0

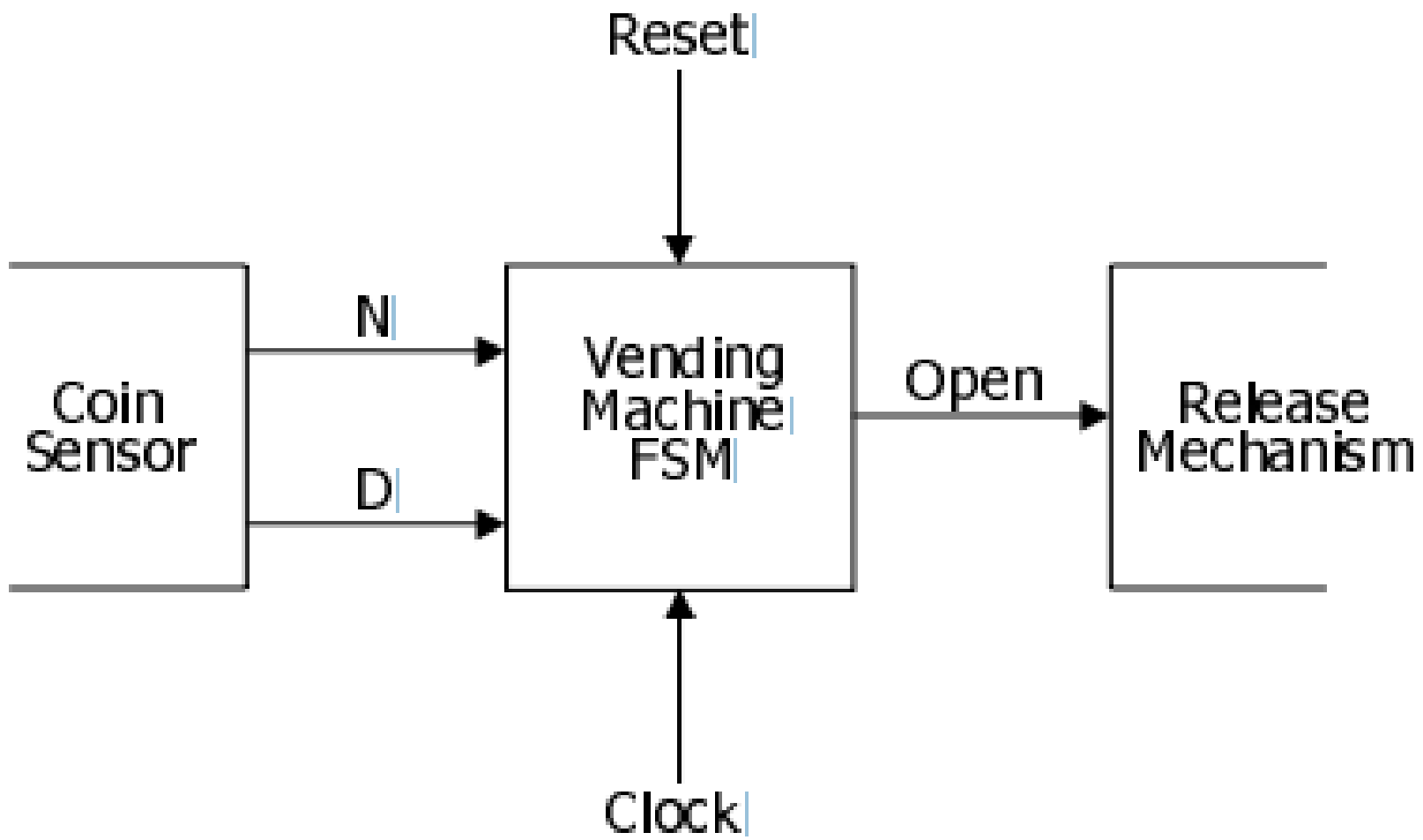
Draw the gate diagram:

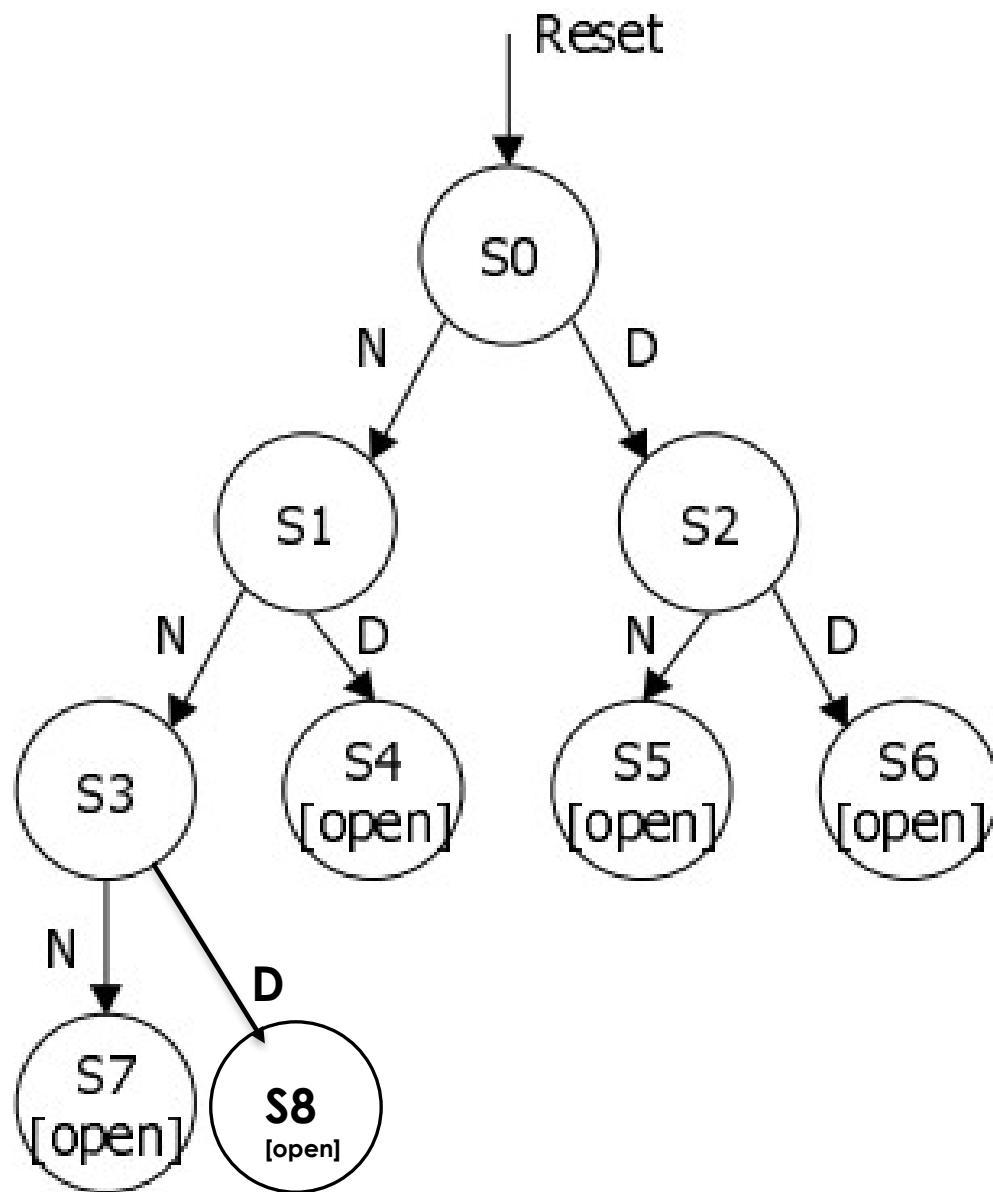


Sequential Circuits Example

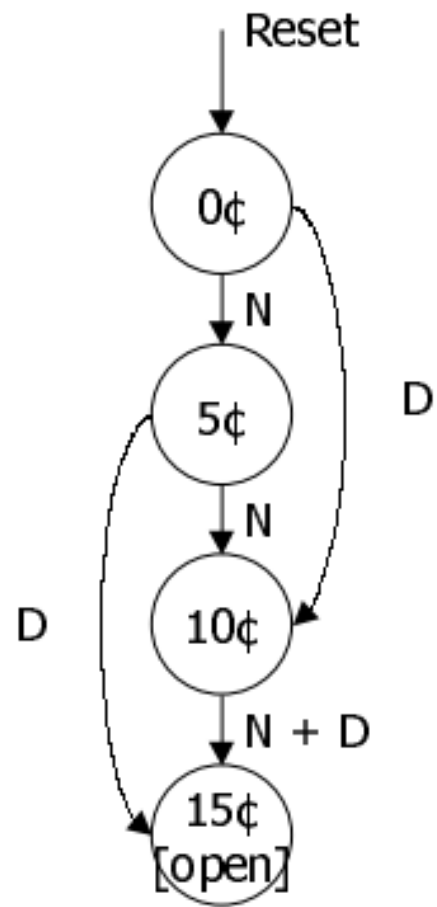
- You have a vending machine with a single coin slot. This machine accepts 5 and 10 cent coins. This machine vends when 15 cents has been put in the machine.
- No change.
- Show the state diagram and state transition table. The state transition table shows the functions to determine the next state and output of the system. Use the states shown below. Possible inputs are N (nickels, 5 cents) or D (dimes, 10 cents).

Q1	Q0	State
0	0	0 cents
0	1	5 cents
1	0	10 cents
1	1	15 cents





- Minimize number of states - reuse states whenever possible



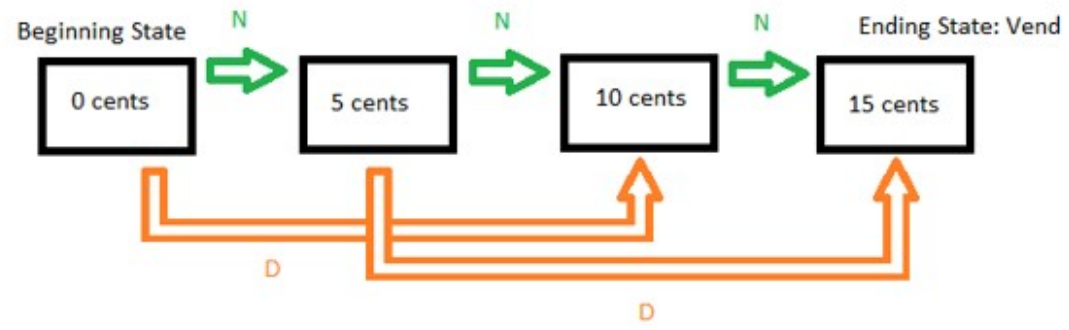
present state	inputs		next state	output open
	D	N		
0¢	0	0	0¢	0
	0	1	5¢	0
	1	0	10¢	0
	1	1	—	—
5¢	0	0	5¢	0
	0	1	10¢	0
	1	0	15¢	0
	1	1	—	—
10¢	0	0	10¢	0
	0	1	15¢	0
	1	0	15¢	0
	1	1	—	—
15¢	—	—	15¢	1

symbolic state table

■ Uniquely Encode States

present state		inputs		next state		output
Q1	Q0	D	N	D1	D0	open
0	0	0	0	0	0	0
		0	1	0	1	0
		1	0	1	0	0
		1	1	—	—	—
0	1	0	0	0	1	0
		0	1	1	0	0
		1	0	1	1	0
		1	1	—	—	—
1	0	0	0	1	0	0
		0	1	1	1	0
		1	0	1	1	0
		1	1	—	—	—
1	1	—	—	1	1	1

State Diagram:

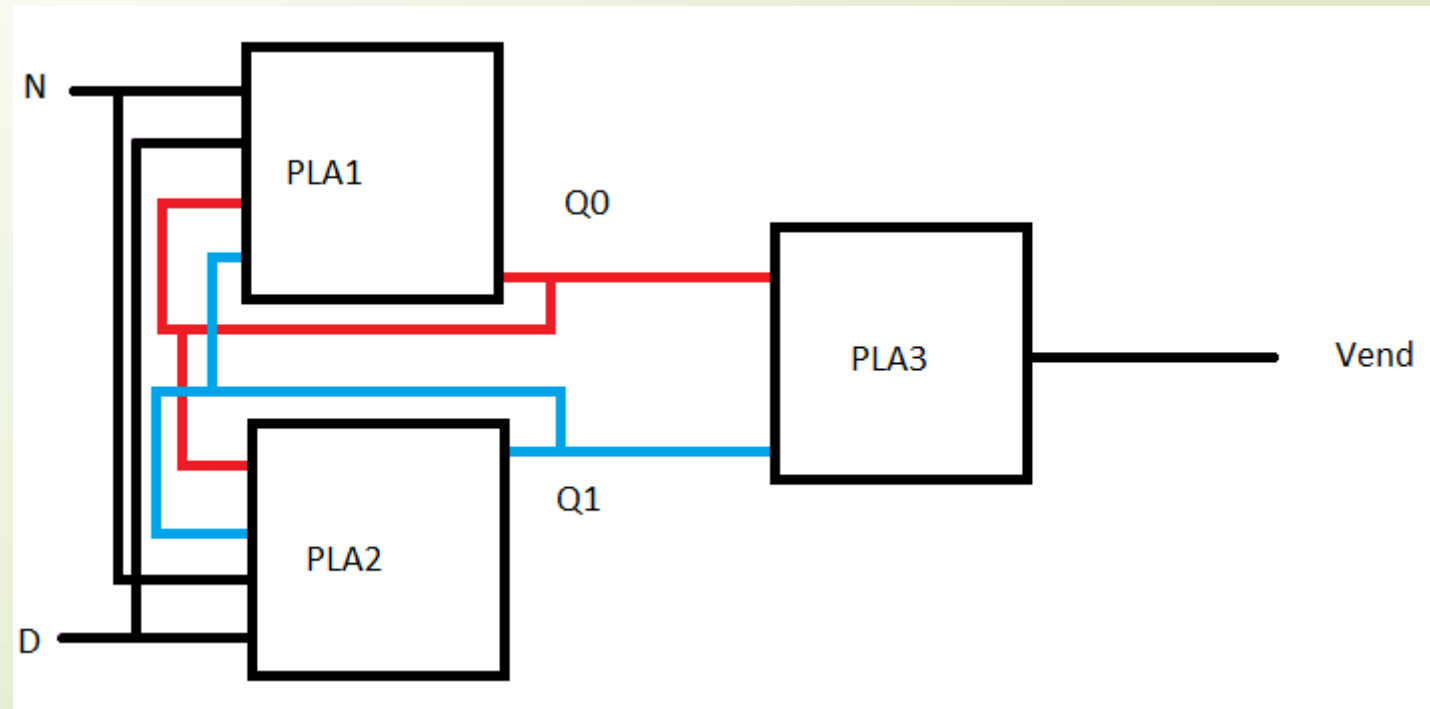


State Transition Table:

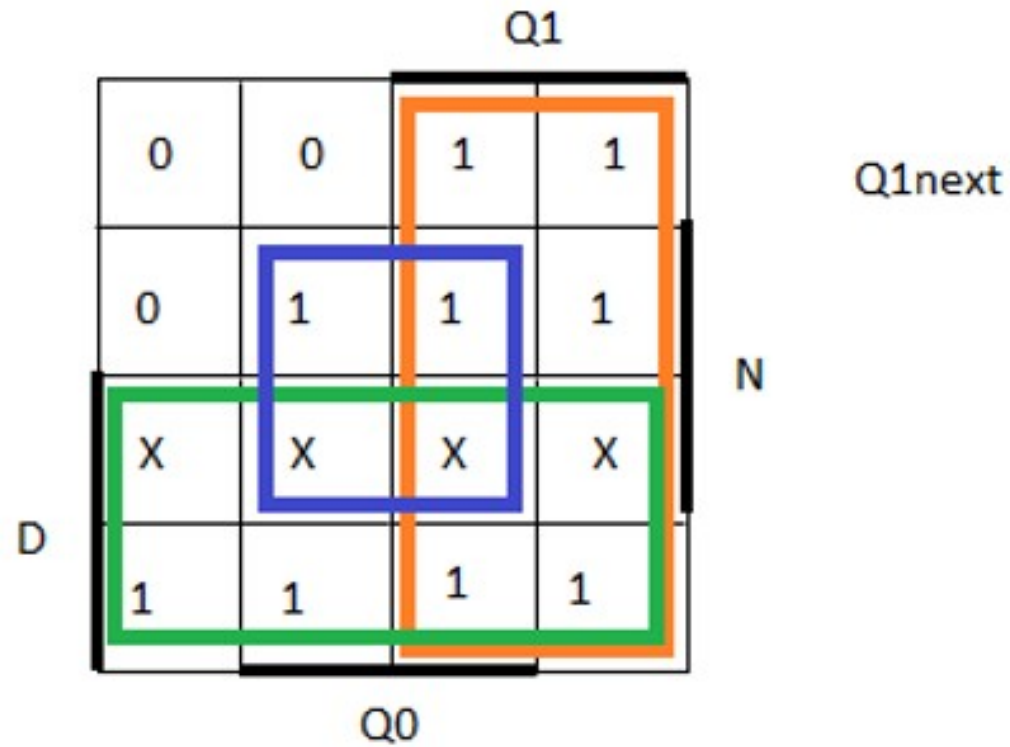
Current States		Input (D) Input (N)		Next State		Output
Q1	Q0	Dime	Nickel	Q1next	Q0next	Vend?
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	1	0	0
0	0	X	X	X	X	X
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	1	1	0
0	1	X	X	X	X	X
1	0	0	0	1	0	0
1	0	0	1	1	1	0
1	0	1	0	1	1	0
1	0	X	X	X	X	X
1	1	0	0	1	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	X	X	X	X	X

Sequential Circuits Continued

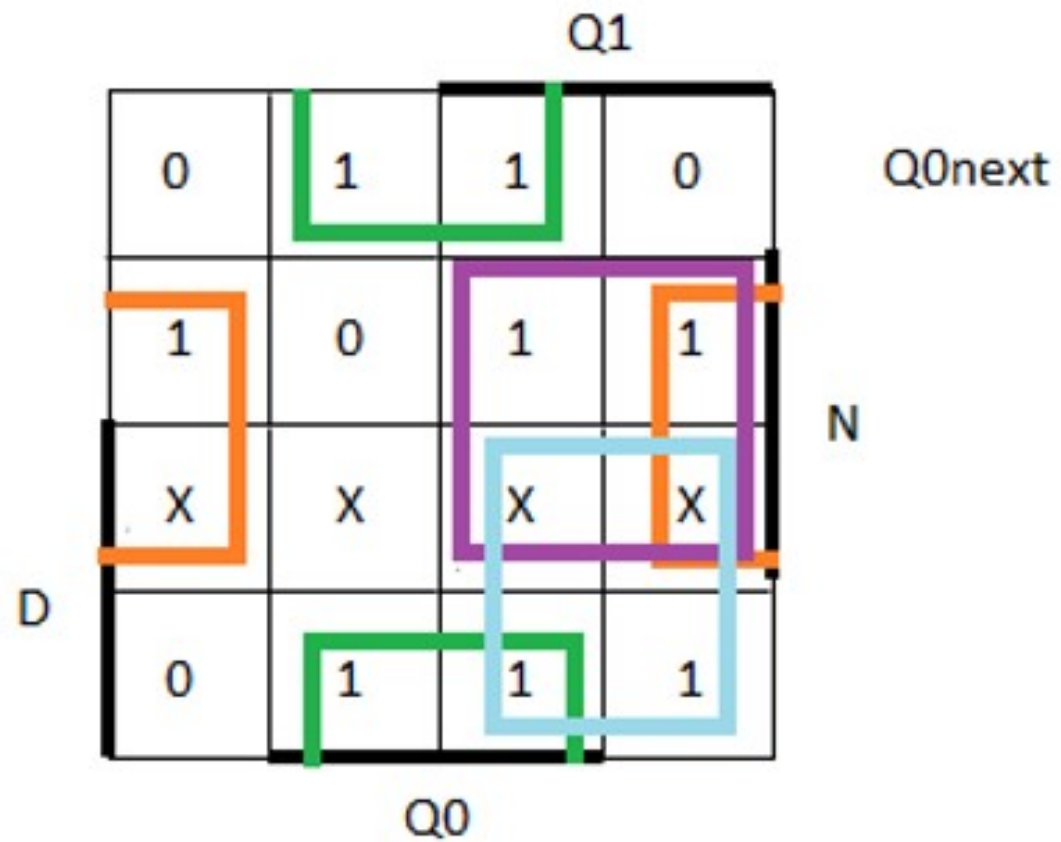
- Reduce the outputs of the state transition table ($Q1_{next}$, $Q0_{next}$, Vend?) to their minimal sum of products form. Show the gate level implementation that fits this circuit.



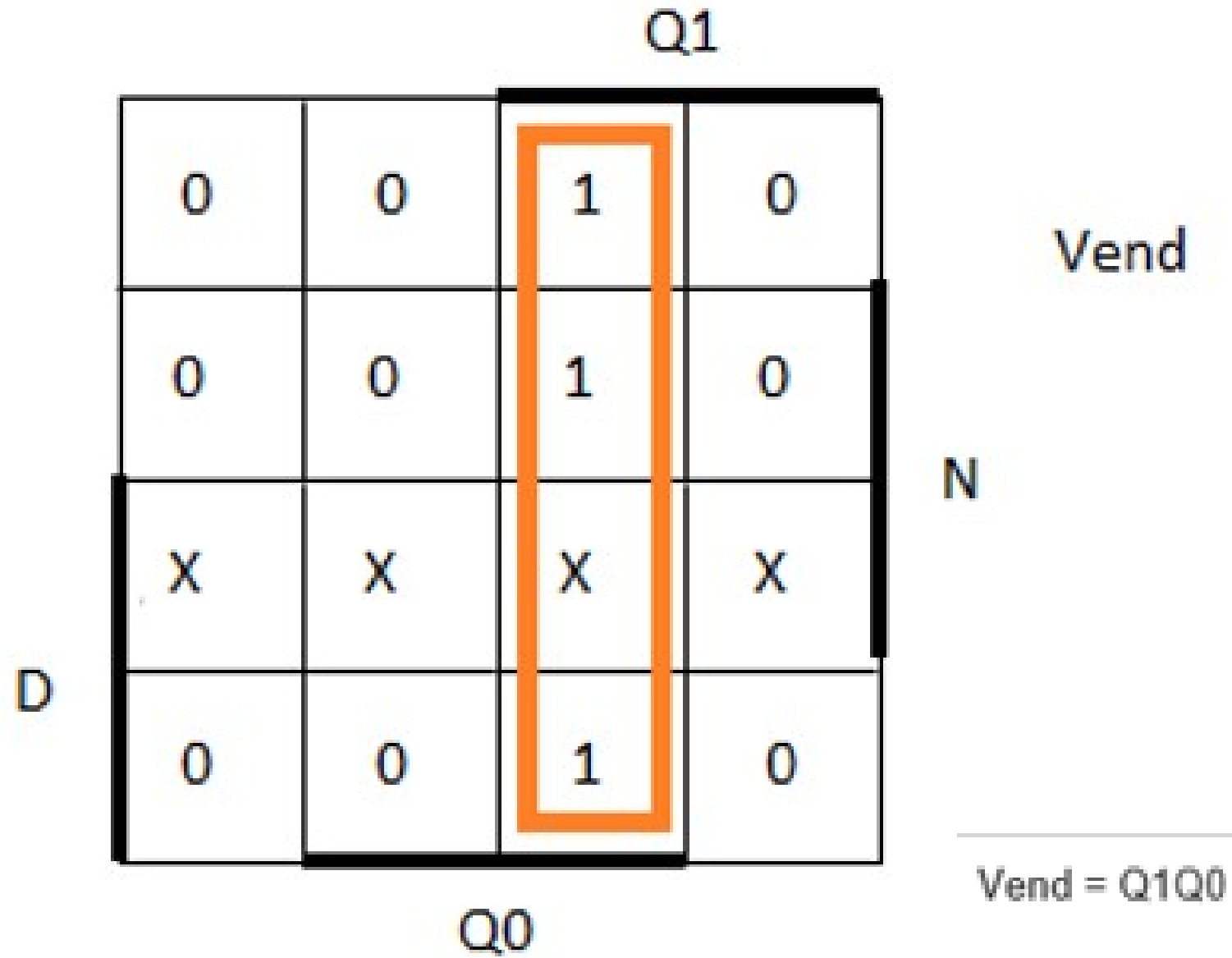
Reduce each output using a Karnaugh map.



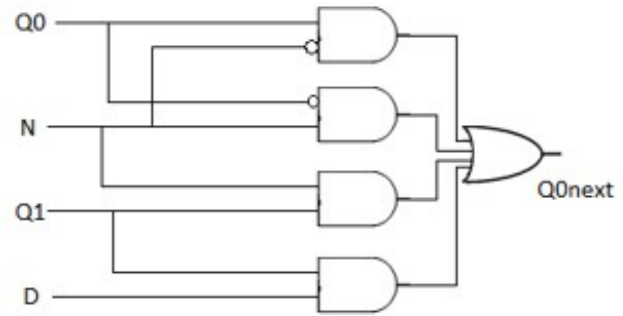
$$Q1_{next} = Q1 + D + Q0N$$



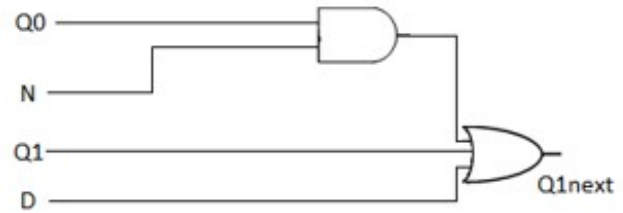
$$Q0_{next} = Q0N' + Q1N + Q1D + Q0'N$$



PLA1 produces Q0next (see circuit above):



PLA2 produces Q1next (see circuit above):



PLA3 produces Vend (see circuit above):

